Parul® University
Vadodara, Gujarat

NAAC GRADE A++

Cryptography
Unit-1

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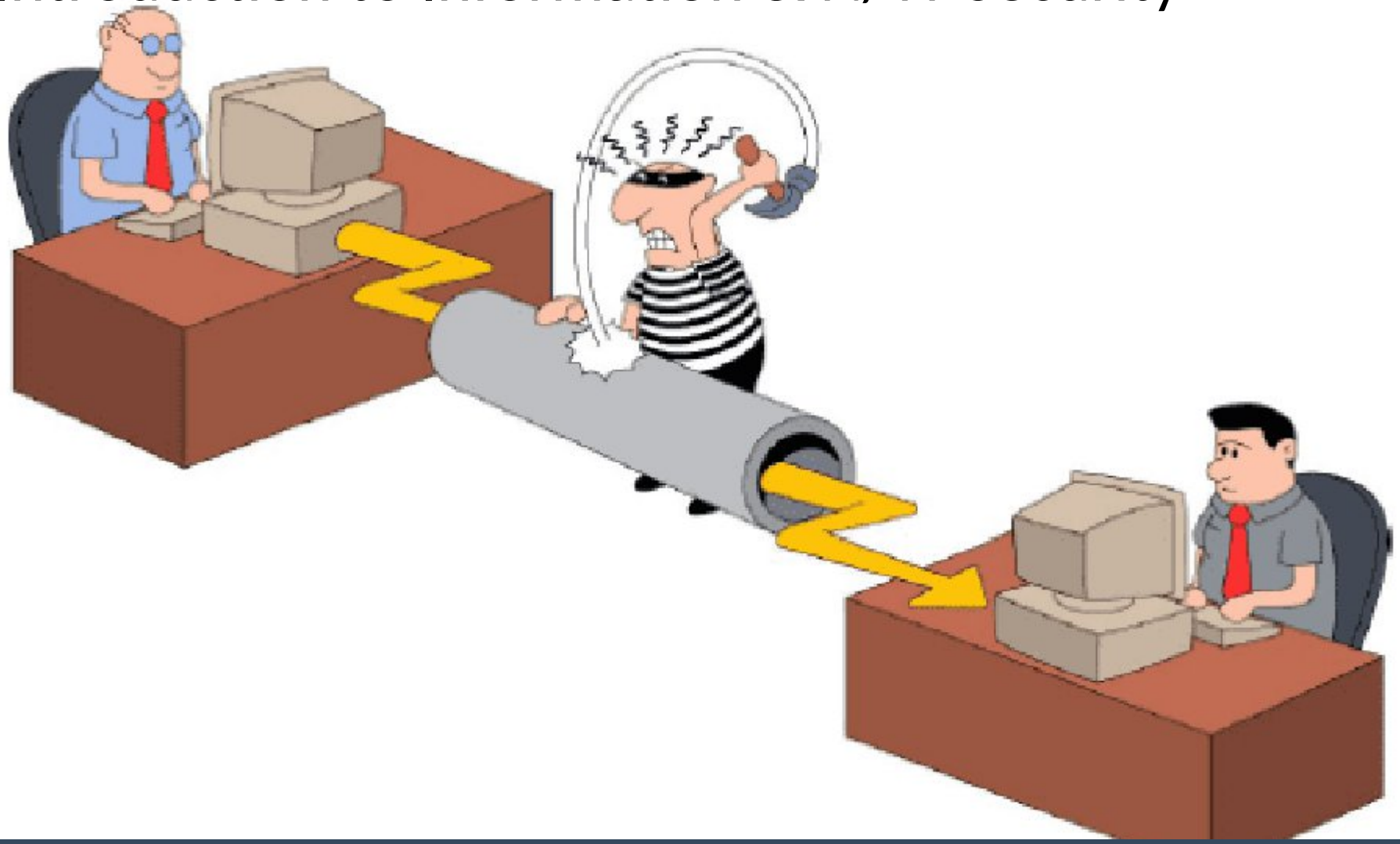
Content

- OSI Security Architecture
- Security Attacks
- Security Services
- Security Mechanism
- Symmetric Cipher Model
- Cryptography
- Cryptanalysis and Attacks
- Substitution and Transposition Techniques

Introduction to Information & N/W Security



OSI Security Architecture

- The OSI (Open Systems Interconnection) security architecture focuses on Security Attacks, Mechanisms, and Services.
- **Security Attack:** Any action that compromises the security of information owned by an organization.
- **Security Mechanism:** A process that is designed to detect, prevent, or recover from a security attack.
- **Security Service:** A communication service that enhances the security of the data processing systems and the information transfers of an organization.

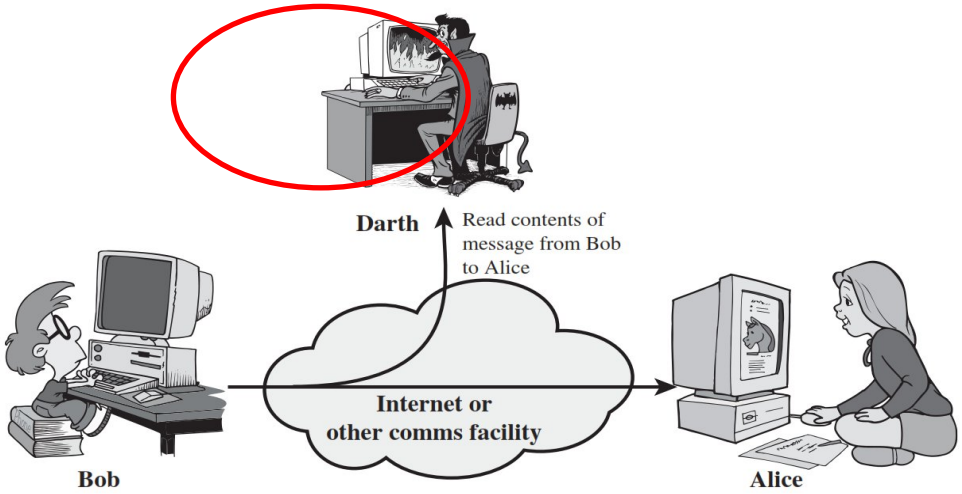
Security Attacks

A **passive attack** attempts to learn or make use of information from the system but does not affect system resources.

1. Release of message contents
2. Traffic analysis

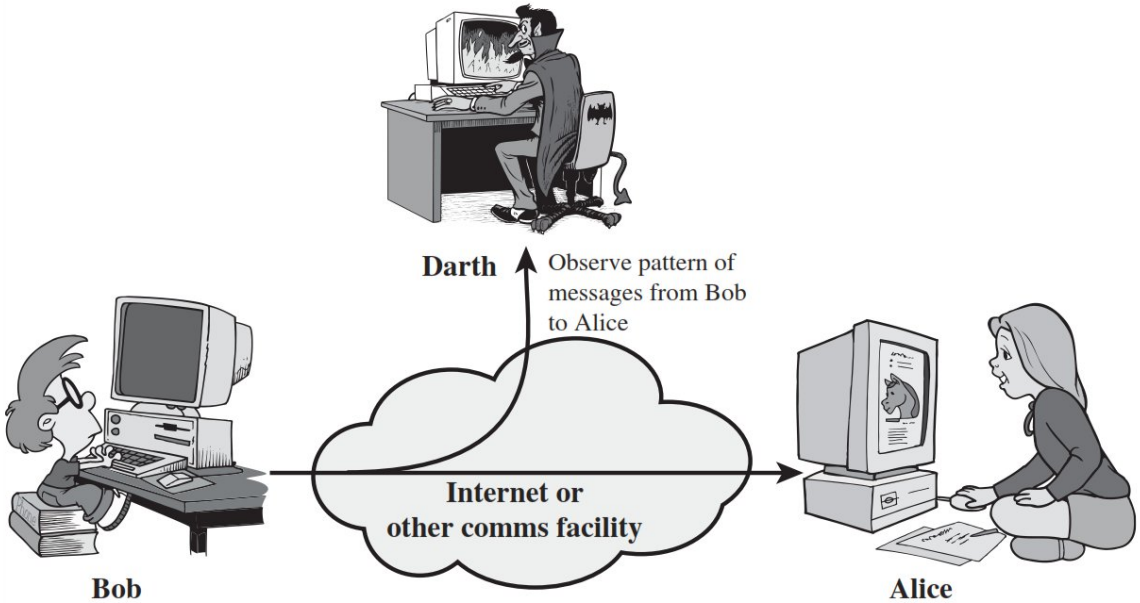
- An **active attack** attempts to alter system resources or affect their operation.
  1. Masquerade
  2. Replay
  3. Modification of messages
  4. Denial of service.

1) Release of message contents (Passive Attack)



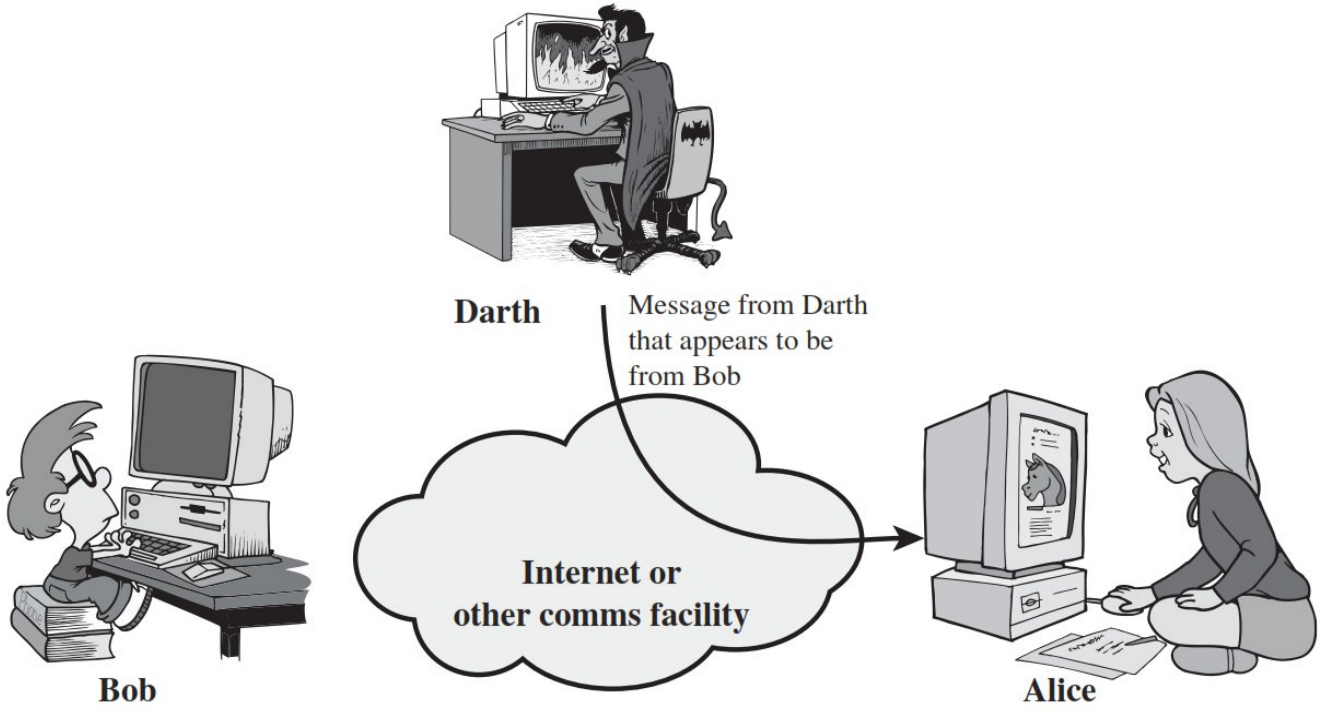
- A telephone conversation, an electronic mail message, and a transferred file may contain sensitive or confidential information.
- We would like to prevent an opponent from learning the contents of these transmissions.

2) Traffic Analysis (Passive Attack)



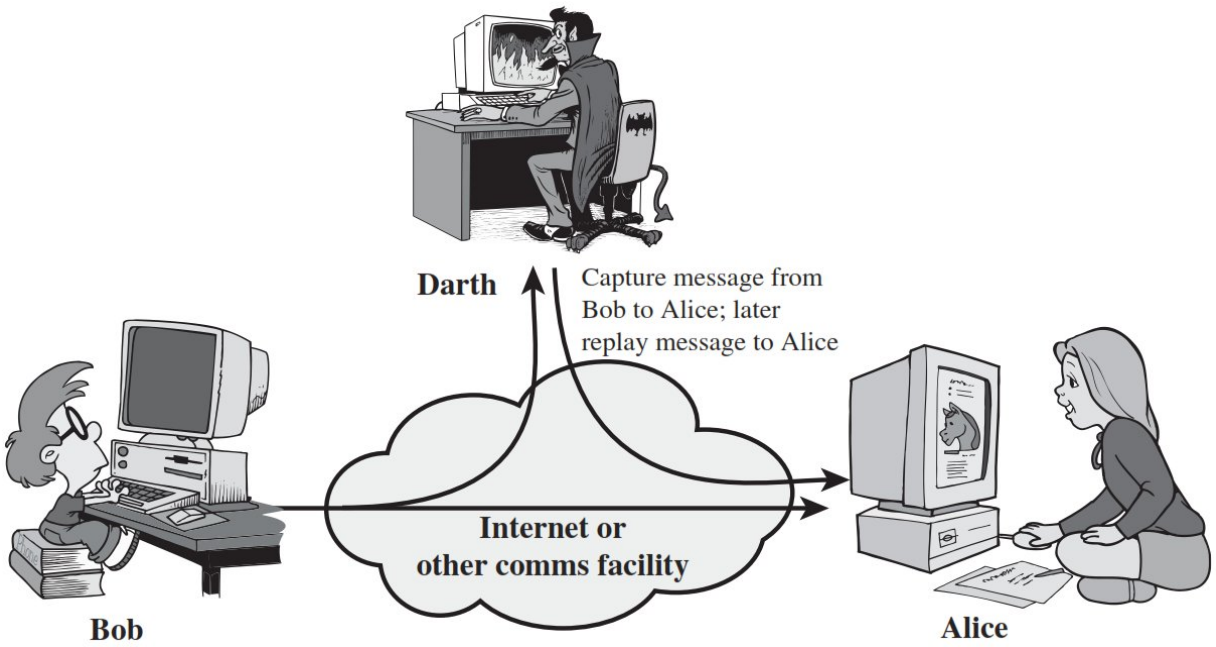
- In such attacks, an adversary, capable of observing network traffic statistics in several different networks, correlates the traffic patterns in these networks.

1) Masquerade Attack (Active Attack)



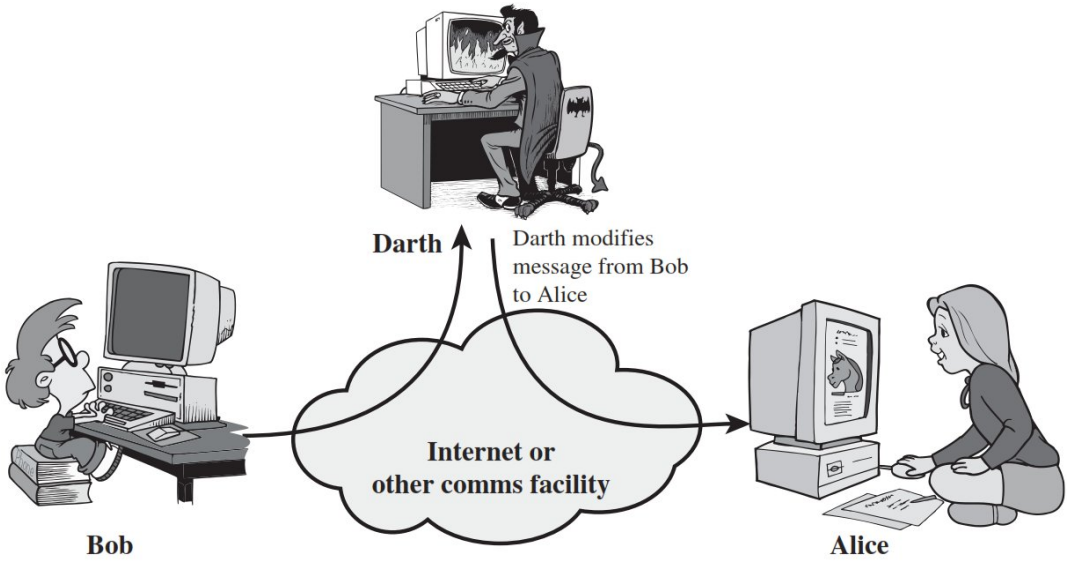
- A **masquerade** takes place when one entity pretends to be a different entity.

2) Replay Attack (Active Attack)



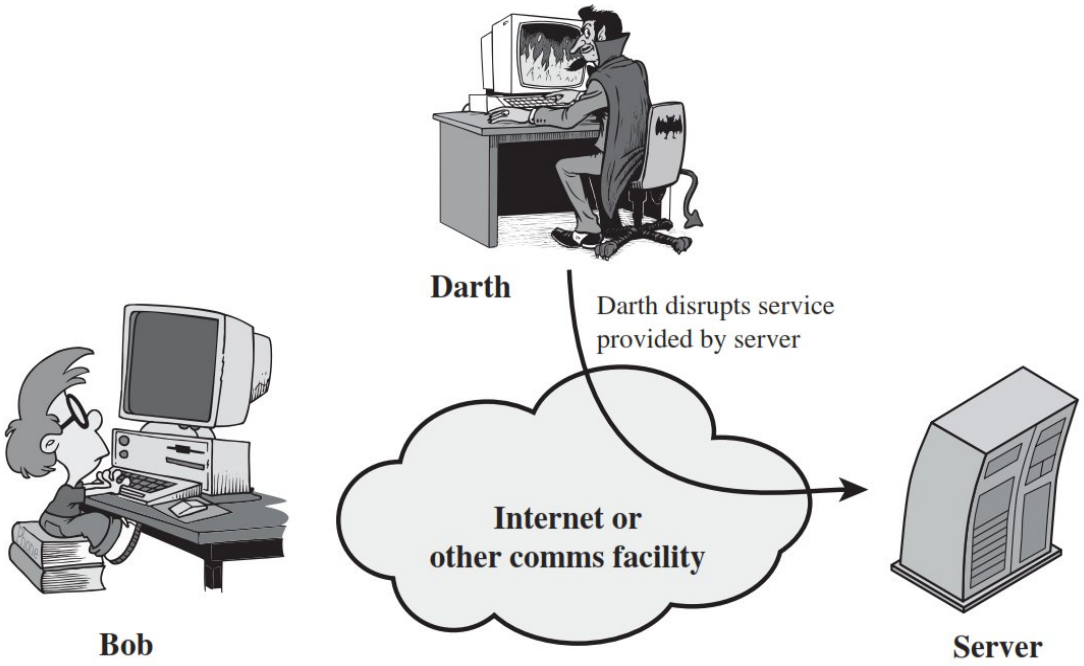
- **Replay attack** involves the passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect.

3) Modification of messages Attack (Active Attack)



- **Modification of messages** simply means that some portion of a legitimate message is altered, or that messages are delayed or reordered, to produce an unauthorized effect.

4) Denial of Service Attack (Active Attack)



- **The denial of service** attack prevents the normal use or management of communications facilities.

Security Services (X.800)

- X.800 standard defines a security service as a service that is provided by a protocol layer of communicating open systems and that ensures security of the systems or of data transfers.

Authentication

- **Authentication** is the assurance that the communicating entity is the one that it claims to be.
  1. **Peer Entity Authentication:** Used in association with a logical connection to provide confidence in the identity of the entities connected.
  2. **Data-Origin Authentication:** In a connectionless transfer, provides assurance that the source of received data is as claimed.

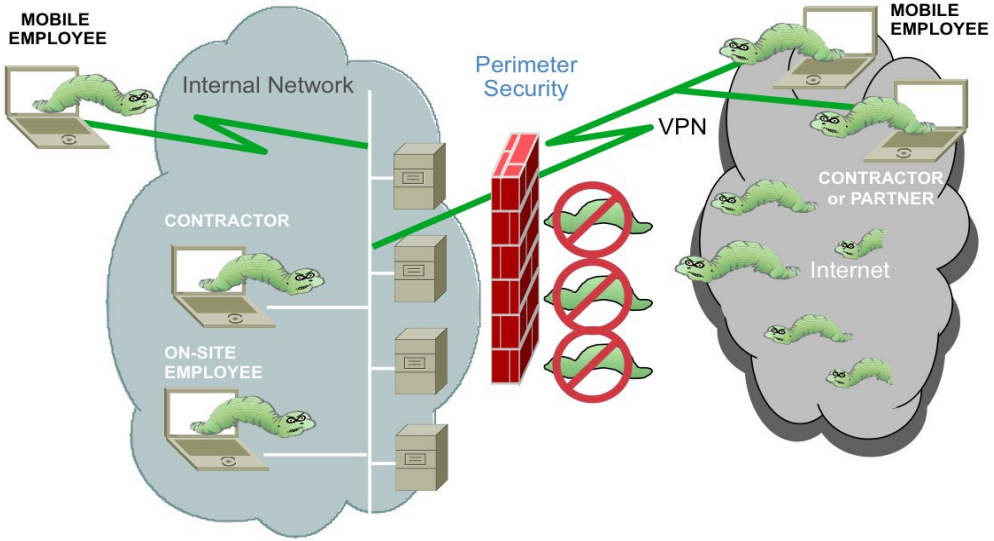
Who you are ?
(biometrics)

Physical authentication
where you are ?

What you know ?
Password
One-time Passwords
Network address

Access Control

- **Access control** is the prevention of unauthorized use of a resource
- This service controls who can have access to a resource, under what conditions access can occur, and what those accessing the resource are allowed to do).



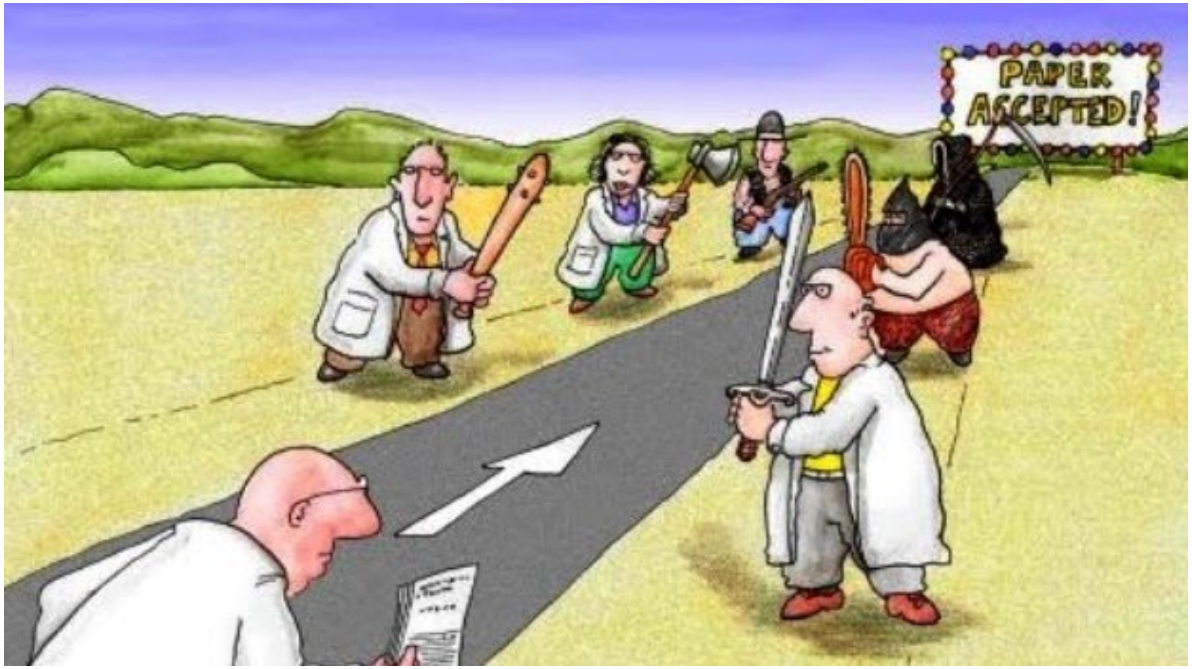
Data Confidentiality

- **Data confidentiality** is the protection of data from unauthorized disclosure.

1. **Connection Confidentiality:** The protection of all user data on a connection.
2. **Connectionless Confidentiality:** The protection of all user data in a single data block.
3. **Selective-Field Confidentiality:** The confidentiality of selected fields within the user data on a connection or in a single data block.
4. **Traffic-Flow Confidentiality:** The protection of the information that might be derived from observation of traffic flows.

Data Integrity

- Data integrity is the assurance that data received are exactly as sent by an authorized entity (i.e., contain no modification, insertion, deletion, or replay).



Data Integrity (Cont…)

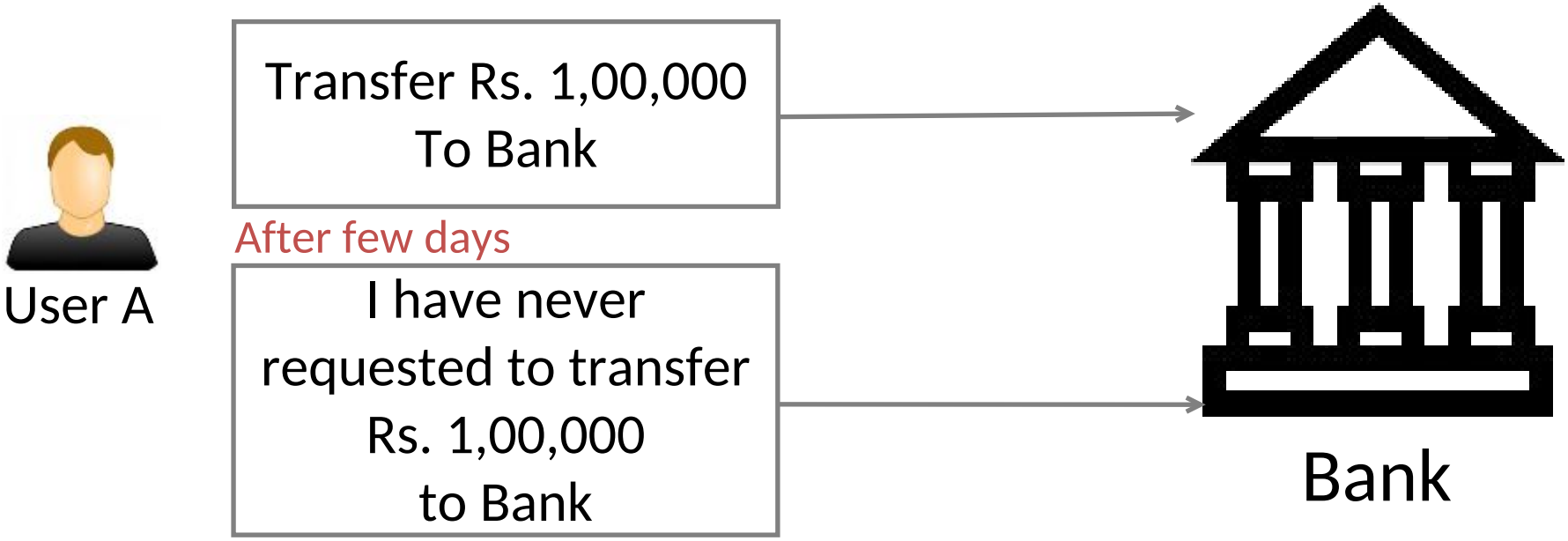
- **Connection Integrity with Recovery:** Provides integrity of all user data on a connection and detects any modification, insertion, deletion, or replay of any data with recovery attempted.
- **Connection Integrity without Recovery:** As above, but provides only detection without recovery.
- **Selective-Field Connection Integrity:** Provides integrity of selected fields within the user data and takes the form of determination of whether the selected fields have been modified, inserted, deleted, or replayed.

Data Integrity (Cont…)

- **Connectionless Integrity:** Provides integrity of a single connectionless data block and may take the form of detection of data modification. Additionally, a limited form of replay detection may be provided.
- **Selective-Field Connectionless Integrity:** Provides integrity of selected fields within a single connectionless data block; takes the form of determination of whether the selected fields have been modified.

Non Repudiation

- **Nonrepudiation** is the assurance that someone cannot deny something.
- Typically, nonrepudiation refers to the ability to ensure that a communication cannot deny the authenticity of their signature on a document or the sending of a message that they originated.



Non Repudiation (Cont…)

- **Nonrepudiation-Origin:** Proof that the message was sent by the specified party.
- **Nonrepudiation-Destination:** Proof that the message was received by the specified party.

Security Mechanisms (X.800)

- **Specific security mechanisms:** Integrated into the appropriate protocol layer in order to provide some of the OSI security services.
- **Pervasive security mechanisms:** Not integrated to any particular OSI security service or protocol layer

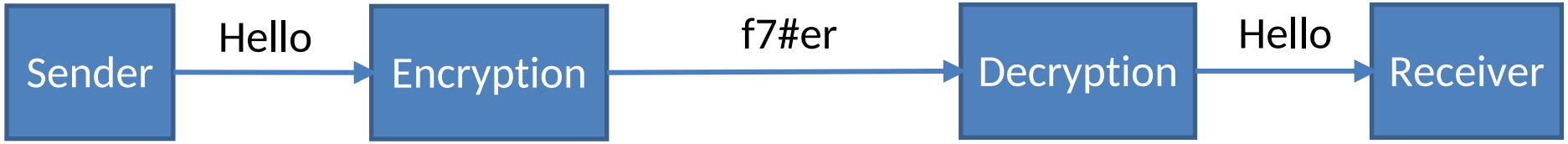
Security Mechanism (Specific Security)

- **Encipherment:** Hiding or covering data using mathematical algorithms.
- **Digital Signature:** The sender can electronically sign the data and the receiver can electronically verify the signature.
- **Access Control:** A variety of mechanisms that enforce access rights to resources.
- **Data Integrity:** A variety of mechanisms used to assure the integrity of a data unit or stream of data units.
- **Authentication Exchange:** Two entities exchange some messages to prove their identity to each other.

Security Mechanism (Specific security)

- **Traffic Padding:** The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts.
- **Routing Control:** Selecting and continuously changing routes between sender and receiver to prevent opponent from eavesdropping.
- **Notarization:** The use of a trusted third party to assure and control the communication.

Encryption and Decryption



Substitution Techniques

- A substitution technique is one in which the letters of plaintext are replaced by other letters or by numbers or symbols.
  1) Caesar Cipher
  2) Monoalphabetic Cipher
  3) Playfair Cipher
  4) Hill Cipher
  5) Polyalphabetic Ciphers
  6) One-Time Pad

1) Caesar Cipher

- The **Caesar cipher** involves replacing each letter of the alphabet with the letter standing three places further down the alphabet.
- In encryption each plaintext letter P, substitute the ciphertext letter C:

$$C = E(k, P) = (P + k) \bmod 26$$

$$C = E(3, P) = (P + 3) \bmod 26$$

- For decr

$$P = D(k, C) = (C - k) \bmod 26$$

- Let us assign a numerical equivalent to each letter

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	1	1	1
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

$$C = E(3, P) = (P + 3) \bmod 26$$

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z
cipher: d e f g h i j k l m n o p q r s t u v w x y z a b c

Example:

Plaintext: THE QUICK BROWN FOX

Ciphertext: WKH TXLFN EURZQ IRA

Brute force attack on Caesar Cipher

- The encryption and decryption algorithms are known.
- There are only 25 keys to try.
- The language of the plaintext is known and easily recognizable.

Ciphertext: ZNK WAOIQ HXUCT LUD

Key	Transformed text
1	YMJ VZNHP GWTBS KTC
2	XLI UYMGO FVSAR JSB
3	WKH TXLFN EURZQ IRA
4	VJG SWKEM DTQYP HQZ
5	UIF RVJDL CSPXOGPY
6	THE QUICK BROWN FOX
7	SGD PTHBJ AQNVM ENW
8	RFC OSGAI ZPMUL DMV
9	QEB NRFZH YOLTK CLU
10	PDA MQEYG XNKSJ BKT
11	OCZ LPDXF WMJRI AJS
12	NBY KOCWE VLIQH ZIR
13	MAX JNBVD UKHPG YHQ

Key	Transformed text
14	LZW IMAUC TJGOF XGP
15	KYV HLZTB SIFNE WFO
16	JXU GKYSA RHEMD VEN
17	IWT FJXRZ QGDLC UDM
18	HVS EIWQY PFCKB TCL
19	GUR DHVPX OEBJA SBK
20	FTQ CGUOW NDAIZ RAJ
21	ESP BFTNV MCZHY QZI
22	DRO AESMU LBYGX PYH
23	CQN ZDRLT KAXFW OXG
24	BPM YCQKS JZWEV NWF
25	AOL XBPJR IYVDU MVE

Substitution Techniques

1) Caesar Cipher
2) **Monoalphabetic Cipher**
3) Playfair Cipher
4) Hill Cipher
5) Polyalphabetic Ciphers
6) One-Time Pad

2) Monoalphabetic Cipher (Simple substitution)

- It is an improvement to the Caesar Cipher.
- Instead of shifting the alphabets by some number, this scheme uses some permutation of the letters in alphabet.
- The sender and the receiver decide on a randomly selected permutation of the letters of the alphabet.
- With 26 letters in alphabet, the possible permutations are 26! which is equal to 4x1026.

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z
cipher: y n l k x b s h m i w d p j r o q v f e a u g t z c

Attack on Monoalphabetic Cipher

- The relative frequencies of the letters in the ciphertext (in percentages) are

P 13.33	H 5.83	F 3.33	B 1.67	C 0.00
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.00
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.00
U 8.33	V 4.17	T 2.50	I 0.83	N 0.00
O 7.50	X 4.17	A 1.67	J 0.83	R 0.00
M 6.67				

Ciphertext:
uzqsovuohxmopvgpozpevsg**zw**szopfpesxudbmetsxaizvuephzhmdzshzowsfpappdtsvpqu**zw**ymxuzuhsxepyepopdzszufpomb**zw**pfupzhmdjudtmohmq

- In our ciphertext, the most common digram is ZW, which appears three times. So equate Z with t, W with h and P with e.
- Now notice that the sequence ZWP appears in the ciphertext, and we can translate that sequence as “the.”

- If the cryptanalyst knows the nature of the plaintext, then the analyst can exploit the regularities of the language.
- The relative frequency of the letters can be determined and compared to a standard frequency distribution for English.
- If the message were long enough, this technique alone might be sufficient, but because this is a relatively short message, we cannot expect an exact match.

Substitution Techniques

1) Caesar Cipher
2) Monoalphabetic Cipher
3) **Playfair Cipher**
4) Hill Cipher
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3) Playfair Cipher

- The Playfair algorithm is based on a 5 × 5 matrix (**key**) of letters.
- The matrix is constructed by filling in the letters of the **keyword** (minus duplicates) from left to right and from top to bottom, and then filling in the remainder of the matrix with the remaining letters in alphabetic order. **The letters I and J count as one letter**.

Example:
Keyword= OCCURRENCE
Plaintext= TALL TREES

Playfair Cipher - Encrypt Plaintext

- Playfair, treats digrams (two letters) in the plaintext as single units and translates these units into ciphertext digrams.
- Make Pairs of letters add filler letter "**X**" if same letter appears in a pair.

Plaintext= TALL TREES

Plaintext= TA LX LT RE ES

- If there is an odd number of letters, then add uncommon letter to complete digram, a X/Z may be added to the last letter.

Playfair Cipher - Encrypt Plaintext

- Map each pair in key matrix

O	C	U	R	E
N	A	B	D	F
G	H	I/J	K	L
M	P	Q	S	T
V	W	X	Y	Z

Plaintext= TA LX LT RE ES

Ciphertext= PF IZ TZ EO RT

- If the letters are on different rows and columns, replace them with the letters on other corner of the same row.
- The order is important - the first letter of the pair should be replaced first.
- For example, using the table above, the letter pair **TA** would be encoded as **PF**.

- For example, using the table above, the letter pair **TA** would be encoded as **PF**.
- § If the letters appear on the same row, replace them with theletters to their immediate right respectively, wrapping around tothe left side of the row if necessary.
- § For example, using the table above, the letter pair **RE** would beencoded as **EO**.

1. Key= " engineering " Plaintext=" test this process "
2. Key= " keyword " Plaintext=" come to the window "
3. Key= " moonmission " Plaintext=" greet "

E N G I R A B C D F H K L M O P Q S T U V W X Y Z	Encrypted Message: pi tu pm gt ue lf gp xg

K E Y W O R D A B C F G H I L M N P Q S T U V X Z	Encrypted Message: lc nk zk vf yo gq ce bw

M O N I S A B C D E F G H K L P Q R T U V W X Y Z	Encrypted Message: hq cz du

Substitution Techniques

1) Caesar Cipher
2) Monoalphabetic Cipher
3) Playfair Cipher
4) **Hill Cipher**
5) Polyalphabetic Ciphers
6) One-Time Pad

4) Hill Cipher

- Hill cipher is based on linear algebra
- Each letter is represented by numbers from 0 to 25 and calculations are done modulo 26.
- Encryption and decryption can be given by the following formula:

Encryption:

$$C = PK \bmod 26$$

Decryption:

$$P = CK^{-1} \bmod 26$$

$$(c_1\ c_2\ c_3) = (p_1\ p_2\ p_3)\begin{pmatrix} k21 & k22 & k23 \\ k31 & k32 & k33 \end{pmatrix}$$

- To encrypt a message using the Hill Cipher we must first turn our keyword and plaintext into a matrix (a 2 x 2 matrix or a 3 x 3 matrix, etc).

Example: Key = "HILL", Plaintext = "EXAM"

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Key Matrix $\begin{pmatrix} H & I \\ L & L \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$

Plaintext $\begin{pmatrix} E \\ X \end{pmatrix} \begin{pmatrix} A \\ M \end{pmatrix} = \begin{pmatrix} 4 \\ 23 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix}$

Key Matrix $= \begin{pmatrix} H & I \\ L & L \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$

Plaintext $\begin{pmatrix} E \\ X \end{pmatrix} \begin{pmatrix} A \\ M \end{pmatrix} = \begin{pmatrix} 4 \\ 23 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix}$

$$C = PK \bmod 26$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 4 \\ 23 \end{pmatrix}$$

$$7 \times 4 + 8 \times 23 = 212$$

$$11 \times 4 + 11 \times 23 = 297$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 4 \\ 23 \end{pmatrix} = \begin{pmatrix} 212 \\ 297 \end{pmatrix}$$

$$\begin{pmatrix} 212 \\ 297 \end{pmatrix} = \begin{pmatrix} 4 \\ 11 \end{pmatrix} \bmod 26 = \begin{pmatrix} E \\ L \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix}$$

$$7 \times 0 + 8 \times 12 = 96$$

$$11 \times 0 + 11 \times 12 = 132$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix} = \begin{pmatrix} 96 \\ 132 \end{pmatrix}$$

$$\begin{pmatrix} 96 \\ 132 \end{pmatrix} = \begin{pmatrix} 18 \\ 2 \end{pmatrix} \bmod 26 = \begin{pmatrix} S \\ C \end{pmatrix}$$

Ciphertext = “ELSC”

Hill Cipher Decryption

$$P = CK^{-1} \bmod 26$$

Step:1 Find Inverse of key matrix

Step:2 Multiply the Multiplicative Inverse of the Determinant by the Adjoin Matrix

Step:3 Multiply inverse key matrix with ciphertext matrix to obtain plaintext matrix

Step: 1 Inverse of key matrix

2 X 2 inverse of matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-cb} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

3 X 3 inverse of matrix

$$A^{-1} = \frac{1}{\text{determinant}(A)} \cdot \text{adjoin}(A)$$

Step: 1 Inverse of key matrix

$$\text{Inverse Key Matrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}^{-1} = \frac{1}{77-88}\begin{pmatrix} 11 & -8 \\ -11 & 7 \end{pmatrix}$$

$$= \frac{1}{-11}\begin{pmatrix} 11 & -8 \\ -11 & 7 \end{pmatrix}$$

$$= \frac{1}{15}\begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix} \bmod 26$$

- -11 mod 26 = 15
- Because, modulo for negative number is = N- (B%N)
  = 26 – (11%26)

Step: 2 Modular (Multiplicative) inverse

- The inverse of a number A is 1/A since A * 1/A = 1
  e.g. the inverse of 5 is 1/5
- In modular arithmetic we do not have a division operation.
- The modular inverse of A (mod C) is A^{-1}
- **$(A * A^{-1}) \equiv 1 \pmod{C}$**

Example:

- The modular inverse of A mod C is the A^{-1} value that makes
  $A * A^{-1} \bmod C = 1$
  A = 3, C = 11
  Since (3*4) mod 11 = 1, **4** is modulo inverse of **3**
  A = 10, C = 17 , $A^{-1} = 12$

- Determinants' multiplicative inverse Modulo 26

Determinant	1	3	5	7	9	11	15	17	19	21	23	25
Inverse Modulo 26	1	9	21	15	3	19	7	23	11	5	17	25

$$= \frac{1}{15}\begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix} \bmod 26$$

- Multiplicative inverse of $\frac{1}{15}$ is 7

Step 2: Multiply with adjoin of matrix

$$= 7\begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix} = \begin{pmatrix} 77 & 126 \\ 105 & 49 \end{pmatrix} = \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \bmod 26$$

$$= \text{thus, if } K = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \text{ then } K^{-1} = \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix}$$

Inverse Key Matrix $= \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix}$ Ciphertext $\begin{pmatrix} E \\ L \end{pmatrix} \begin{pmatrix} S \\ C \end{pmatrix} = \begin{pmatrix} 4 \\ 11 \end{pmatrix} \begin{pmatrix} 18 \\ 2 \end{pmatrix}$

$$P = CK^{-1} \bmod 26$$

$$\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 4 \\ 11 \end{pmatrix}$$

$$25 \times 4 + 22 \times 11 = 342$$

$$1 \times 4 + 23 \times 11 = 257$$

$$\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 4 \\ 11 \end{pmatrix} = \begin{pmatrix} 342 \\ 257 \end{pmatrix}$$

$$\begin{pmatrix} 342 \\ 257 \end{pmatrix} = \begin{pmatrix} 4 \\ 23 \end{pmatrix} \bmod 26 = \begin{pmatrix} E \\ X \end{pmatrix}$$

$$\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 18 \\ 2 \end{pmatrix}$$

$$25 \times 18 + 22 \times 2 = 494$$

$$1 \times 18 + 23 \times 2 = 64$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix} = \begin{pmatrix} 494 \\ 64 \end{pmatrix}$$

$$\begin{pmatrix} 494 \\ 64 \end{pmatrix} = \begin{pmatrix} 0 \\ 12 \end{pmatrix} \bmod 26 = \begin{pmatrix} A \\ M \end{pmatrix}$$

Plaintext = “EXAM”

Substitution Techniques

1) Caesar Cipher
2) Monoalphabetic Cipher
3) Playfair Cipher
4) Hill Cipher
5) **Polyalphabetic Ciphers**
6) One-Time Pad

5) Polyalphabetic Cipher

- Monoalphabetic cipher encoded using only one fixed alphabet
- **Polyalphabetic cipher** is a substitution cipher in which the cipher alphabet for the plain alphabet may be different at different places during the encryption process.
  1. **Vigenere cipher**
  2. **Vernam cipher**

PT = **HELLO**

KEY = **GM**

CT = **NQRXU**

Key

Plaintext

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Vigenere Cipher

Keyword : **DECEPTIVE**

Key : **DECEPTIVE**

Plaintext : **WEAREDISCOVEREDSAVEYOURSELF**

Ciphertext : **ZICVTWQNGRZGVTWAVZHCQYGLMGJ**

$$C = (P_1 + K_1, P_2 + K_2, \ldots P_m + K_m) \bmod 26$$

$$P = (C_1 - K_1, C_2 - K_2, \ldots C_m - K_m) \bmod 26$$

An analyst looking at only the ciphertext would detect the repeated sequences VTW at a displacement of 9 and make the assumption that the keyword is either three or nine letters in length.

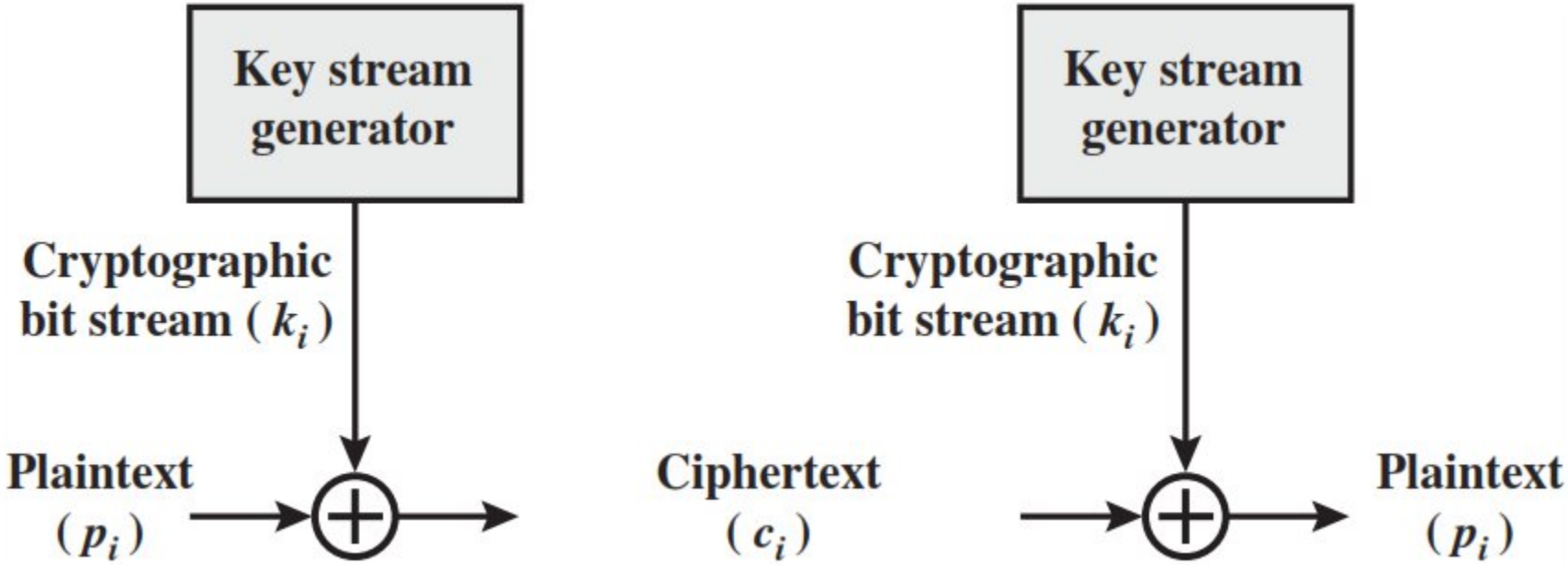
Keyword : **DECEPTIVE**

Key : **DECEPTIVE**

Plaintext : **WEAREDISCOVEREDSAVEYOURSELF**

This system is referred as an **autokey system**

- The ciphertext is generated by applying the logical XOR operation to the individual bits of plaintext and the key stream.



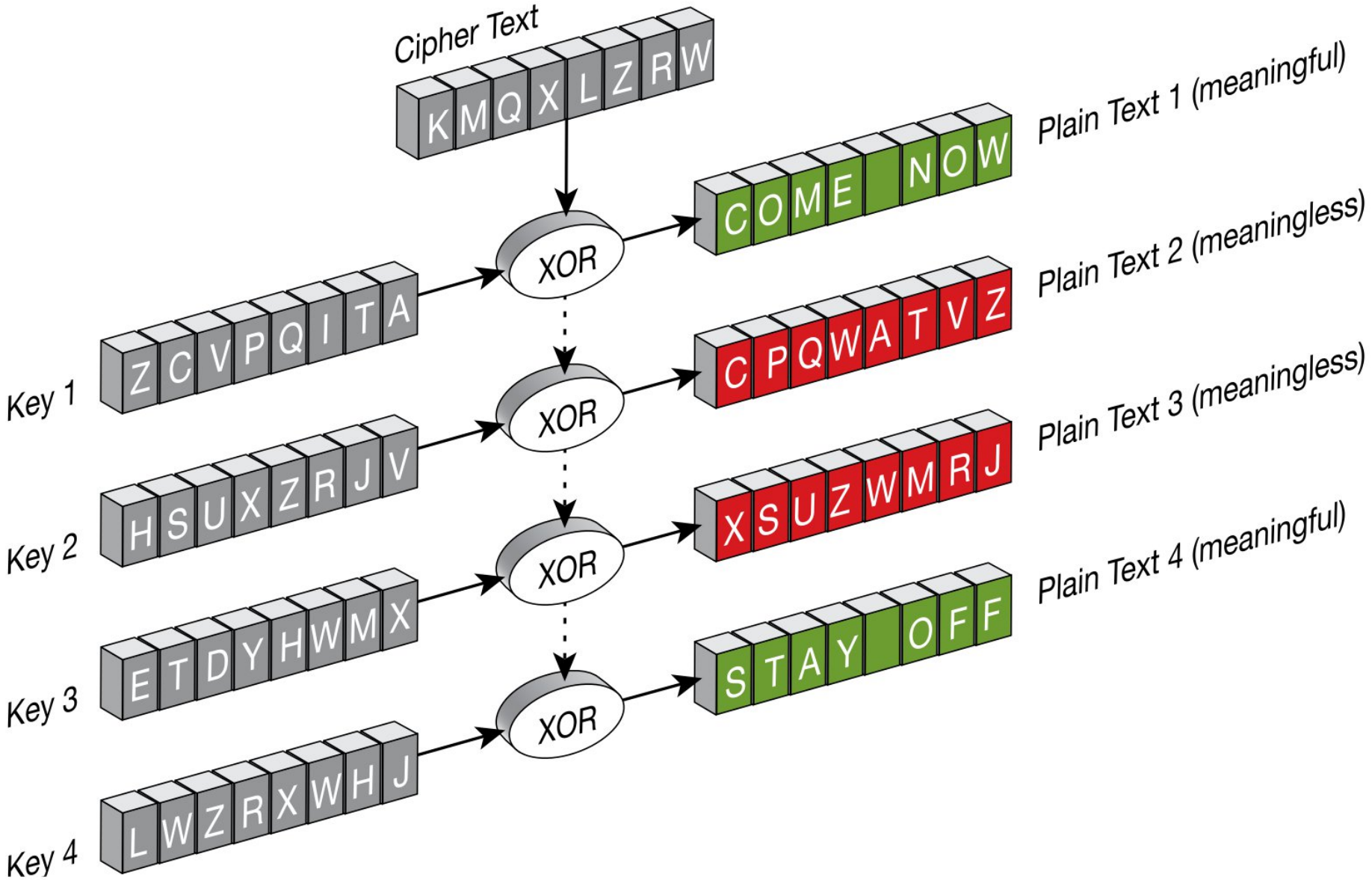
Substitution Techniques

1) Caesar Cipher
2) Monoalphabetic Cipher
3) Playfair Cipher
4) Hill Cipher
5) Polyalphabetic Ciphers
6) **One-Time Pad**

One time pad

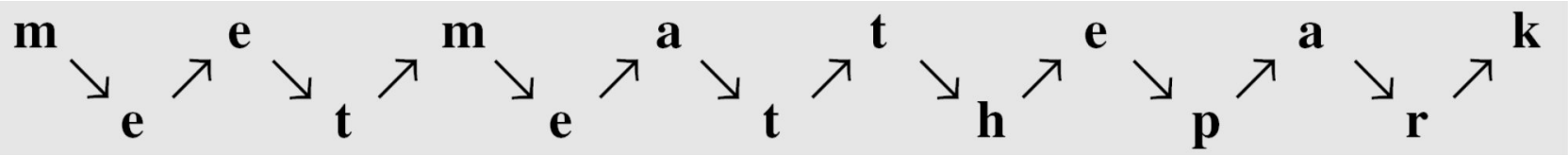
- The one-time pad, which is a provably secure cryptosystem, was developed by Gilbert Vernam in 1918.
- The message is represented as a binary string (a sequence of 0's and 1's using a coding mechanism such as ASCII coding.
- **The key is a truly random sequence of 0's and 1's of the same length as the message.**

- message ='IF'
- then its ASCII code =(1001001 1000110)
- key = (1010110 0110001)
- *Encryption*:
  - 1001001 1000110 plaintext
  - 1010110 0110001 key
  - 0011111 1110110 ciphertext


Cipher Text
K M Q X L Z R W
Key 1
Z C V P Q I T A
Key 2
H S U X Z R J V
Key 3
E T D Y H W M X
Key 4
L W Z R X W H J
XOR
XOR
XOR
XOR
C O M E N O W
C P Q W A T V Z
X S U Z W M R J
S T A Y O F F
Plain Text 1 (meaningful)
Plain Text 2 (meaningless)
Plain Text 3 (meaningless)
Plain Text 4 (meaningful)

Transposition Techniques

- A transposition cipher does not substitute one symbol for another, instead it changes the location of the symbols.
- The simplest such cipher is the **rail fence technique**, in which the plaintext is written down as a sequence of diagonals and then read off as a sequence of rows.
- For example, to send the message **"Meet me at the park"** to Bob, Alice writes



- She then creates the ciphertext "MEMATEAKETETHPR".

Rail fence technique

- A more complex scheme is to write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns.
- The order of the columns then becomes the key to the algorithm.

```
Key:          4 3 1 2 5 6 7
Plaintext:    a t t a c k p
              o s t p o n e
              d u n t i l t
              w o a m x y z
Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ
```